

UNIT - IV

Test of hypothesis for small samples:-

- * student t-test
- * Hypothesis testing containing one mean & two mean.

T-distribution:-

- * It is used for testing of hypothesis when the sample size is small ($n < 30$) and population S.D is unknown.

→ If $\{x_1, x_2, x_3, \dots, x_n\}$ be any random sample of size n drawn from normal population with μ , variance σ^2 , then the test statistics $t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ when $\bar{x}$ = sample mean

$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ is an unbiased estimator of σ^2

→ The test statistics $t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ is a random variable having the t-distribution with $v = n - 1$ degree of freedom and with probability density function.

$f(t) = \frac{y_0}{\sqrt{v}} \left(1 + \frac{t^2}{v}\right)^{-\frac{v+1}{2}}$ when $v = n - 1$ degree of freedom and y_0 is a constant is obtained by the total probability

$$\int_{-\infty}^{\infty} f(t) dt = 1$$

→ This is known as t distribution (or) student-t distribution.

* Test of hypothesis of single mean :- (small samples)

(a) The null hypothesis is no diff between means $H_0: \bar{x} = \mu$

(b) The alternate hypothesis is both are not equal $\bar{x} \neq \mu$

(c) The LOS $\alpha = 10\%$ or 5% or 1%

(d) The test statistics under H_0 is $t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

(e) If $|t_{cal}| < t_{table}$, we accept H_0
If $|t_{cal}| > t_{table}$, we reject H_0

(1) The sample of 26 bulbs given a mean life of 990 hrs with a S.D of 20 hrs. The manufacturer claims that the mean life of bulbs is 1000 hrs is the sample upto the standard or not.

soln:-

$\bar{x} = 990$ hrs, $n = 26$

$\sigma = 20$ hrs, $\mu = 1000$ hrs.

$$v = n - 1 = 25$$

(a) H_0 : The sample is upto std

(b) H_1 : The sample is not upto std.

(c) The level of $\alpha = 5\%$

(d) The test statistic under H_0 is

$$t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{990 - 1000}{20/\sqrt{26}} = \frac{-10 \times \sqrt{26}}{20} = -2.549 \approx -2.55$$

(e) The 5% table val with 25 dof is 1.708

$$|t_{cal}| > t_{table}$$

$\therefore$ We reject H_0

2) The avg breaking strength of steel rods is specified 18.5 thousand pounds. To test this sample of 14 rods for tested, the mean and S.D are 17.85 and 1.955 respectively is the result of the experiment significant.

Soln
 1) H_0 : The result is significant

2) H_1 : The result is not significant.

3) The level of $\alpha = 5\%$

4) The test statistics under H_0 is

$$t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{17.85 - 18.5}{1.955/\sqrt{14}} = \frac{-0.65 \times \sqrt{14}}{1.955}$$

$$= -1.244$$

5) The 5% table val with $v = 13$ dof

$$is = 1.771$$

$\therefore |t_{cal}| < t_{table}$; we accept H_0 .

3) The random sample of size 16 from a normal population with mean 53 and the sum of squares of deviations from the mean 150. The sample be regarded has taken from the populations having 56 has the mean obtain 95% confidence limits from mean of population?

Soln

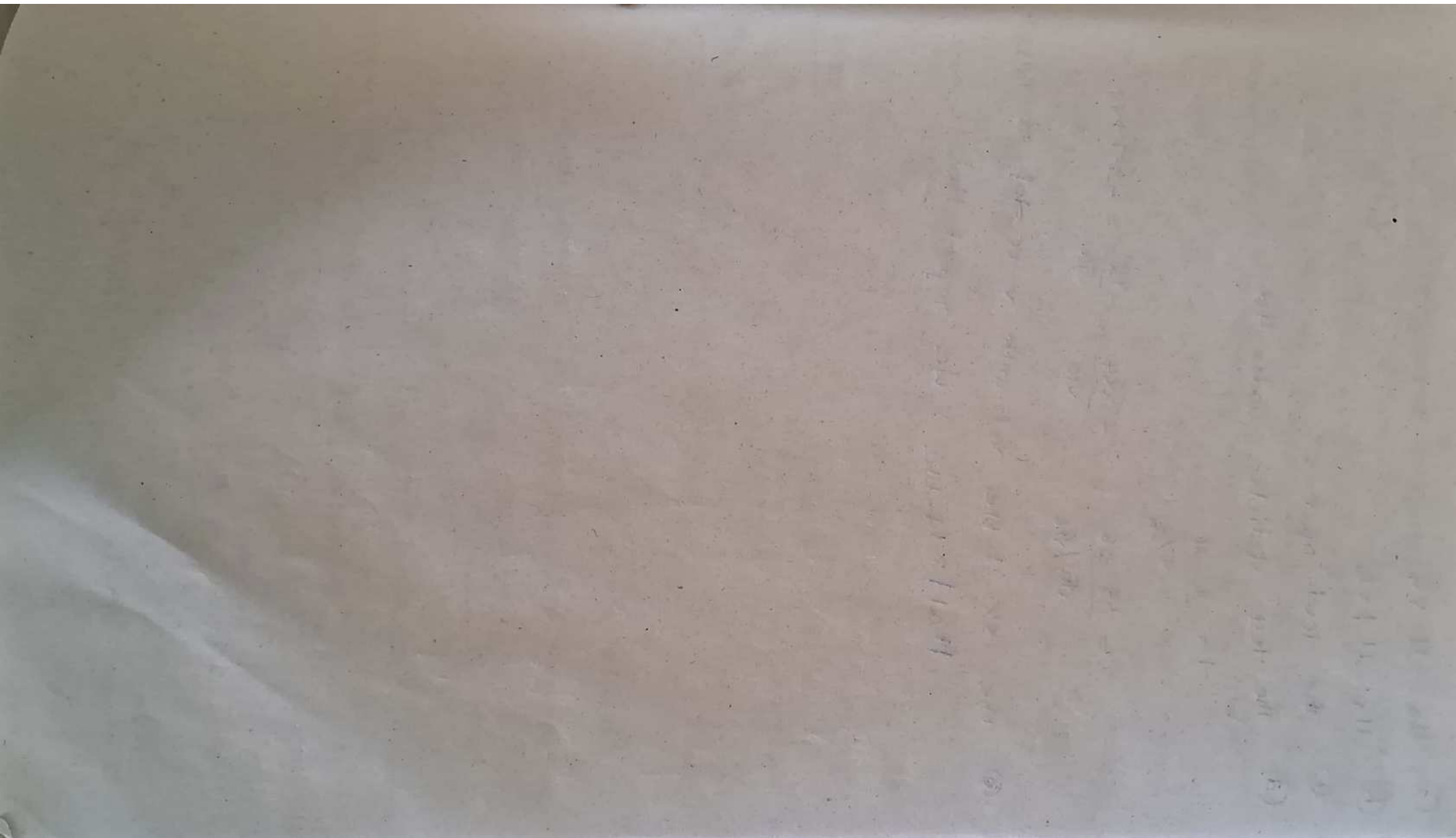
$$n = 16, \bar{x} = 53$$

$$\sum (x_i - \bar{x})^2 = 150$$

$$\mu = 56$$

$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2 = \frac{150}{15}$$

$$= 10$$



(a) $H_0: \mu = 56$

(b) $H_1: \mu \neq 56$

(c) The level of $\alpha = 5\%$

(d) The test statistic under H_0

$$t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$= \frac{53 - 56}{\sqrt{10}/\sqrt{16}} = -\frac{3 \times 4}{\sqrt{10}} = -\frac{12}{\sqrt{10}} = -3.7947$$

(e) The 5% table val with $v=15$ dof is $t_{5\%} = 1.753$

$|t_{cal}| > t_{table}$; we reject H_0 .

② Test of significance difference of means (small samples):

Let $\bar{x}$ & $\bar{y}$ be the sample means of two independent samples of sizes n_1 & n_2 rly.

(Both n_1 & $n_2 < 30$) drawn from two normal populations having means μ_1 & μ_2 to test whether the two sample means are equal or not equal.

step 1:- The null hypothesis

H_0 : The two means are equal

$$\mu_1 = \mu_2$$

step 2:-

H_1 : The two means are not equal

$$\mu_1 \neq \mu_2$$

step 3:- The LOS $\alpha = 5\%$ (or) 1%

If $\sigma_1^2 = \sigma_2^2 = \sigma^2$ then an unbiased estimate s^2 of the common variance of σ^2 is given by $s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$ where s_1^2 & s_2^2 are two sample variances

step 4:- The test-statistic under H_0 is

$$t = \frac{\bar{x} - \bar{y}}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

step 5:- If $|t_{cal}| < t_{table}$ we accept H_0 with
 degree of freedom $(n_1 + n_2 - 2)$

If $|t_{\text{cal}}| > t_{\text{table}}$ we reject H_0
with degree of freedom $(n_1 + n_2 - 2)$

Ex ① Samples of 2 types of electric bulbs were tested for length of life and following data were obtained

Type 1	Type 2
$n_1 = 8$	$n_2 = 7$
$\bar{x} = 1284$ hrs	$\bar{y} = 1056$ hrs
$S_1 = 36$ hrs	$S_2 = 40$ hrs

Is the difference in the means sufficient to variant.

Variant that type 1 is more superior to

type 2 regarding the length of life.

Soln:

a) $H_0: \mu_1 = \mu_2$

b) $H_1: \mu_1 \neq \mu_2$

c) $\alpha = 5\%$

d) $s^2 = \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2 - 2} = \frac{8 \times (36)^2 + 7 \times (40)^2}{8 + 7 - 2}$

$s^2 = 1659.07$

$s = \sqrt{1659}$

$= 40.73$

d) Test statistic

$t = \frac{1284 - 1056}{40.73 \sqrt{\frac{1}{8} + \frac{1}{7}}} = \frac{228}{40.73 \times 0.517} = 10.8$

~~$|t_{cal}| > t_{table}$ we reject H_0 of dof 13.~~

The 5% tabulated with $(8+1-2) = 13$ dof is

$|t_{cal}| > t_{table}$ we reject H_0 . 1.771

- ② Two horses A & B were tested according to the time (in secs) to run a particular track with the following results.

A	28	30	32	33	33	29	34
B	29	30	30	24	27	29	-

Test whether the two horses have the same running capacity.

Soln Given that

∴ The $n_1 = 7$ $n_2 = 6$

$$\bar{x} = \frac{28+30+32+33+33+29+34}{7}$$

$$= \frac{219}{7} = 31.28$$

$$\bar{y} = \frac{29+30+30+24+27+29}{6} = \frac{169}{6} = 28.16$$

$$S^2 = \frac{1}{n_1+n_2-2} \left[\sum (x_i - \bar{x})^2 + (y_i - \bar{y})^2 \right]$$

$$= \frac{1}{7+6-2} \left[(28-31.28)^2 + (30-31.28)^2 + (33-31.28)^2 + (33-31.28)^2 + (29-31.28)^2 + (34-31.28)^2 + (29-28.16)^2 + (30-28.16)^2 + (30-28.16)^2 + (24-28.16)^2 + (27-28.16)^2 + (29-28.16)^2 \right]$$

$$s^2 = 5.23 \Rightarrow s = \sqrt{5.23} = 2.29$$

(a) $H_0: \mu_1 = \mu_2$

(b) $H_a: \mu_1 \neq \mu_2$

(c) $\alpha = 5\%$

(d) $t = \frac{\bar{x} - \bar{y}}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{31.2 - 28.1}{2.29 \sqrt{\frac{1}{7} + \frac{1}{6}}} = 2.43$

(e) the 5% tabulated with 11 dof is

$$= 1.796$$

$$|t_{cal}| > t_{table}$$

we reject H_0 .

(3) To examine the hypothesis that the husbands are more intelligent than wives and investigator took a sample of 10 couples and observed them a test which measures the IQ level. The results are given the following table:

Test the hypothesis at 5% LOS, $t_{cal} = 1.19$
 $t_{tab} = 1.71$

H	117	105	97	105	123	109	86	78	103	107
W	106	104	87	104	116	95	90	69	108	85

The first of these is the fact that the
 world is not a uniform whole, but is
 divided into many different parts, each
 with its own special characteristics.
 These differences are due to a variety of
 causes, including climate, soil, and
 human activity. The second point is that
 the world is not a static entity, but is
 constantly changing. This is due to the
 fact that the world is a dynamic system,
 in which all parts are interrelated and
 influence each other. The third point is
 that the world is not a simple system,
 but is a complex one, with many
 different levels of organization. This
 complexity is due to the fact that the
 world is a system of systems, each of
 which is itself a complex system.

② F - Test of Significance:-

When testing the significances of two samples we assume that the 2 samples came from the same population with equal variances. If the variances of the population are not equal, A significant difference in the means may arise. Hence before we apply the t-test for the difference of two means, we have to test the equality of population variances using F-test of significance.

- If s_1^2 & s_2^2 are the 2 sample variances of sizes n_1 & n_2 r/y then the population variances are given by

$$\sigma_1^2 \text{ (or) } \hat{\sigma}_1^2 = \frac{n_1 s_1^2}{n_1 - 1}$$

$$\sigma_2^2 \text{ (or) } \hat{\sigma}_2^2 = \frac{n_2 s_2^2}{n_2 - 1}$$

- The quantities $n_1 - 1$ & $n_2 - 1$ are called degrees of freedom of these estimates. we want to test if these estimates $\hat{\sigma}_1^2$ & $\hat{\sigma}_2^2$ are significantly different or not

step 1:- The null hypothesis $H_0: \sigma_1^2 = \sigma_2^2$

step 2:- $H_1: \sigma_1^2 \neq \sigma_2^2$

step 3: $\alpha = 5\% \text{ (or)} 1\%$

step 4: The test statistic under H_0 is

$$\boxed{\begin{aligned} F &= \frac{S_1^2}{S_2^2} \quad (S_1^2 > S_2^2) \\ f &= \frac{S_2^2}{S_1^2} \quad (S_2^2 > S_1^2) \end{aligned}}$$

$$S_1^2 = \frac{1}{n_1 - 1} \sum (x_i - \bar{x})^2$$

$$S_2^2 = \frac{1}{n_2 - 1} \sum (y_i - \bar{y})^2$$

step 5:- If $F_{cal} < F_{table}$ we accept
 $(n_1 - 1, n_2 - 1)$ dof
 $v_1 \quad v_2$

ex ① Two Random samples reveal the following

results			
Sample	sizes	Means	sum of squares of deviations from means
1	10	15	90
2	12	14	108

$$s_1^2 = \frac{1}{n_1-1} \sum (x_i - \bar{x})^2 = \frac{90}{10-1} = 10$$

$$s_2^2 = \frac{1}{n_2-1} \sum (y_i - \bar{y})^2 = \frac{108}{12-1} = \frac{108}{11} = 9.82$$

(a) $H_0: s_1^2 = s_2^2$

(b) $H_1: s_1^2 \neq s_2^2$

(c) $\alpha = 5\%$

(d) Test statistic $F = \frac{10}{9.82} = 1.018$

(e) $F_{cal} = 1.018$
 $F_{table} = 2.90$
 $(1.018 < 2.90)$
 we accept H_0 .

25/04/2025

(2) The time taken by workers in performing a job by method-I & method-II is given below. Do the data show that the variances of time distribution from population from which the samples are drawn do not differ significantly.

Method I	20	16	26	27	23	22	
Method II	27	33	42	35	32	34	38

Soln - Given that

The sample sizes $n_1 = 6$ $n_2 = 7$

$$\bar{x} = \frac{\sum x_i}{n} = \frac{20+16+26+27+23+22}{6} = \frac{134}{6} = 22.33$$

$$\bar{y} = \frac{27+33+42+35+32+34+38}{7} = 34.4$$

$$s_1^2 = \frac{1}{n_1-1} \sum (x_i - \bar{x})^2$$

$$= \frac{1}{5} [(20-22.3)^2 + \dots + (22-22.3)^2]$$

$$= \frac{1}{5} [5.29 + 39.69 + 13.69 + 22.09 + 0.49 + 0.09]$$

$$= \frac{1}{5} [81.34]$$

$$= 16.26$$

$$s_2^2 = \frac{1}{n_2-1} \sum (y_i - \bar{y})^2 = \frac{1}{6} [(27-34.4)^2 + \dots + (38-34.4)^2]$$

$$= 22.29$$

$$(1) H_0: s_1^2 = s_2^2$$

$$(2) H_1: s_1^2 \neq s_2^2$$

$$(3) \alpha = 5\% \text{ LOS with } (6, 5) \text{ dof}$$

$$(4) F_{cal} = \frac{s_2^2}{s_1^2} = \frac{22.29}{16.26} = 1.37$$

$$(5) \text{ The } 5\% \text{ table value with } (6, 5) \text{ dof} \\ u = 4.95$$

$F_{cal} < F_{tab}$, we accept H_0 .

- ③ The nicotine contains in mg in 2 samples of tobacco were found to be as follows

I	24	27	26	21	25	—
II	27	30	28	31	22	36.

Test the hypothesis that the two samples came from the same normal population.

$$\begin{cases} s_1^2 = 5.3 \\ s_2^2 = 21.6 \end{cases}$$

$$f_{\text{cal}} = 4.075$$

$$f_{\text{tab}} = 6.26$$

We accept H_0 .

The Chi-square Distribution (χ^2)

- When a fair coin is tossed 100 times, the theoretical consideration leads us to expect 50 heads and 50 tails. But in practice these results are rarely achieved i.e. the results obtained in an experiment do not agree exactly with the theoretical results.
- The magnitude of disturbance b/w the theory and experimental observations is given by the Quantity χ^2 . If $\chi^2 = 0$ the observed and expected values completely coincide.
- If χ^2 value increases the disturbance b/w observed and expected frequencies increases.
- Thus χ^2 affords a measure of the correspondence b/w theory and experimental frequencies.

Defn If a set of events $A_1, A_2, A_3, \dots, A_n$ are observed to occur with frequencies $O_1, O_2, \dots, O_n$ respectively and according to probability rules

$A_1, A_2, \dots, A_n$ are expected to occur with frequencies $E_1, E_2, \dots, E_n$ respectively. If

O_i ($i=1$ to n) is a set of observed values

E_i ($i=1$ to n) is a set of Expected values

then χ^2 is defined as

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} \text{ with } (n-1) \text{ dof}$$

* $\rightarrow \chi^2$ is used to test whether the differences b/w the observed and expected frequencies are significant

$\rightarrow \chi^2$ distribution is an important continuously probability distribution and is used in both large and small sample test. In χ^2 test χ^2 distribution is used to test the goodness of fit to test the independence of attributes.

(i) χ^2 test as a test of Goodness of fit
→ we use this test to decide the disturbancy between theory and experimental values is significant or not.

① The Null hypothesis is H_0 : There is no significant difference b/w observed & expected value ($O_i = E_i$)

② The alternative hypothesis is H_1 : There is a difference b/w ($O_i \neq E_i$)

③ The LOS $\alpha = 5\%$ or 1%

④ The test statistic under H_0 is

$$\chi^2_{cal} = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

⑤ if $\chi^2_{cal} < \chi^2_{table}$ we accept H_0 with $(n-1)$ dof

if $\chi^2_{cal} > \chi^2_{table}$ we reject H_0 with $(n-1)$ dof

S-A

* Conditions of Validity of χ^2 test:-

→ following are the conditions which should be satisfied before χ^2 test can be applied.

(i) The sample observations should be independent

(ii) The total frequency $[N > 50]$

(iii) The constraints w in the self frequencies if any are linear.

No theoretical frequency should be less than 10.

- ① the no. of automobile accidents per week in a certain community are as follows
 12, 8, 20, 2, 14, 10, 15, 06, 9 & 4 are these frequencies in agreement with the belief that accident conditions were the same during this 10 week period.

Soln. Expected frequency of accidents each week

$$= \frac{N}{10} = \frac{100}{10} = 10$$

② H_0 : The accident conditions were the same during the 10 week period. or.

③ H_1 : the accident conditions are different

O_i	E_i	$O_i - E_i$	$(O_i - E_i)^2$
12	10	2	4
8	10	-2	4
20	10	10	100
2	10	-8	64
14	10	4	16
10	10	0	0
15	10	5	25
06	10	-4	16
9	10	-1	1
4	10	-6	36

$$\sum (O_i - E_i)^2 = 266$$

$$\chi^2_{cal} = \frac{266}{10} = 26.6$$

The 5% val with 9 dof $\alpha = 16.919$

$\chi^2_{cal} > \chi^2_{table} \therefore$ we reject H_0 .

② A sample analysis of examination results of 500 students was made it was found that 220 students had failed, 170 students had secured a 3rd class, 90 students in second class, and 20 students got 1st class. Do these figures with the general examination results with ratio 4:3:2:1 for the various category χ^2 -

Solⁿ: ① H_0 : The observed results commensurate with the general examination results. ($O_i = E_i$)

The expected frequency are in the ratio 4:3:2:1

Total frequency = 500. If divide the total frequency 500 in the ratio 4:3:2:1

we get the expected frequency 200, 150, 100, 50.

② H_1 : $O_i \neq E_i$

O_i	E_i	$O_i - E_i$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
220	200	20	400	$400/200 = 2$
170	150	20	400	$400/150 = 2.6$
90	100	-10	100	$100/100 = 1$
20	50	-30	900	$900/50 = 18$

$$\chi^2_{cal} = \frac{\sum (O_i - E_i)^2}{E_i}$$

The 5% table val with 3 dof

$$= 7.815$$

$$= 23.667$$

$$\chi^2_{cal} > \chi^2_{table}$$

$\therefore$ we reject H_0 .

24/10/25
 ⑤ A pair of dice are thrown 360 times and the frequency of each sum and the frequency of each sum is given in the following table.

sum	2	3	4	5	6	7	8	9	10	11	12
frequency	8	24	35	37	44	65	51	42	26	14	14

could say that the dice are fair on the basis of χ^2 distribution at 5% of LOS

soln/
 (a) H_0 : The dice are fair

(b) H_1 : The dice are unfair

(c) The LOS $\alpha = 5\%$

The probabilities getting a sum 2, 3, 4, 5, 6, 7, 8, ..., 12 and the corresponding expected frequencies are given in the following table.

$x = x_i$	2	3	4	5	6	7	8	9	10	11	12
$P(X = x_i)$	$1/36$	$2/36$	$3/36$	$4/36$	$5/36$	$6/36$	$7/36$	$4/36$	$3/36$	$2/36$	$1/36$
E_i	10	20	30	40	50	60	50	40	30	20	10

$\frac{1}{36} \times 360$

O_i	E_i	$(O_i - E_i)$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
8	10	-2	4	$4/10 = 0.4$
24	20	4	16	$16/20 = 0.8$
35	30	5	25	$25/30 = 0.83$
37	40	-3	9	$9/40 = 0.25$
44	50	-6	36	$36/50 = 0.72$
65	60	5	25	$25/60 = 0.41$
51	50	1	1	$1/50 = 0.02$
42	40	2	4	$4/40 = 0.1$
26	30	-4	16	$16/30 = 0.53$
14	20	-6	36	$36/20 = 1.8$
14	10	+4	16	$16/10 = 1.6$

$$\textcircled{d} \chi^2_{\text{cal}} = \frac{\sum (O_i - E_i)^2}{E_i}$$

$$= 7.445$$

$\textcircled{e}$ 5% Table val with $(12-1)$ dof is ~~18.307~~ 19.675

$$\chi^2_{\text{cal}} < \chi^2_{\text{table}}$$

we Accept H_0 .

4. Test of hypothesis of independence of Attributes:-

→ An attribute means a quality or characteristics

Example:- Drinking, blindness, honesty, beauty etc...

→ An attribute may be marked its position in a number of a given population.

Let the observations be classified according to 2 attributes and the frequencies O_{ij} in the different character categories, be shown in a 2 way table called contingency table

- we have to test on the basis of the self frequencies whether the two attributes are independent or not.

- we take the Null hypothesis H_0 : there is no association b/w the two attributes i.e we assume that

$\textcircled{a}$ H_0 : The two attributes are independent

$\textcircled{b}$ H_1 : The two attributes are not //

$\textcircled{c}$ $\alpha = 5\%$ or 1%

— The expected frequency of any cell is

$$= \frac{\text{row total} \times \text{column total}}{\text{Grand total}}$$

④ The test statistic under H_0 is

$$\chi^2_{\text{cal}} = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

with dof $(r-1)(c-1)$ dof

⑤ If the calculated value is less than tabulated value then we accept H_0 i.e. the two attributes are independent

— If the cal. value is greater than tabulated value we reject H_0 i.e. the two attributes are not independent. 29/04/25

① The following table gives the classification of 100 workers according to the males and females and nature of work. Test whether the nature of 'work' is independent of the sex of males and females or work

	stable	unstable	Total
Males	40	20	60
Females	10	30	40
Total	50	50	100

① The H_0 : The nature of work is independent of Males and female workers

② H_1 : The two attributes are not independent

③ The LOS = 5%

The expected frequencies

$$E_{11} = \frac{60 \times 50}{100} = 30$$

$$E_{12} = \frac{60 \times 50}{100} = 30$$

$$E_{21} = \frac{40 \times 50}{100} = 20$$

$$E_{22} = \frac{40 \times 50}{100} = 20$$

O_{ij}	E_{ij}	$(O_{ij} - E_{ij})$	$(O_{ij} - E_{ij})^2$	$\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$
40	30	10	100	$100/30 = 3.33$
20	30	-10	100	$100/30 = 3.33$
10	20	-10	100	$100/20 = 5$
30	20	10	100	$100/20 = 5$
				$\chi^2_{cal} = 16.66$

④ $\chi^2_{cal} = 16.66$

⑤ 5% Table value with 1 dof is 3.841

$$\chi^2_{cal} > \chi^2_{table}$$

We reject H_0 .

$\therefore$ The two attributes are not independent

2. Two researchers adopted different sampling techniques while investigating some group of students to find the no. of students failing into different intelligence level. The results are given in following table.

Researches	Below average	Avg	Above avg	Genius	Total
X	86	60	44	10	200
Y	40	33	25	2	100
Total	126	93	69	12	300

→ Could you say that the sampling techniques adopted by two researchers are significantly different at 5% LOS.

- ① H_0 : There is no significant difference b/w 2 researchers
- ② H_1 : There is significant difference b/w 2 researchers
- ③ LOS = 5%

The expected frequencies

$$E_{11} = \frac{200 \times 126}{300} = 84$$

$$E_{12} = \frac{200 \times 93}{300} = 62$$

$$E_{13} = \frac{200 \times 69}{300} = 46$$

$$E_{14} = \frac{200 \times 12}{300} = 8$$

$$E_{21} = \frac{100 \times 126}{300} = 42$$

$$E_{22} = \frac{100 \times 93}{300} = 31$$

$$E_{23} = \frac{100 \times 69}{300} = 23$$

$$E_{24} = \frac{100 \times 12}{300} = 4$$

O_{ij}	E_{ij}	$O_{ij} - E_{ij}$	$(O_{ij} - E_{ij})^2$	$\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$
86	84	2	4	$4/84 = 0.04$
60	62	-2	4	$4/62 = 0.06$
44	46	-2	4	$4/46 = 0.08$
10	8	2	4	$4/8 = 0.5$
40	42	-2	4	$4/42 = 0.09$
33	31	2	4	$4/31 = 0.12$
25	23	2	4	$4/23 = 0.17$
2	4	-2	4	$4/4 = 1$

$$\chi^2_{cal} = 2.097$$

⑤ 5% Table value with 3 dof
is 7.82

$$\chi^2_{cal} < \chi^2_{table}$$

$\therefore$ we accept H_0 .

③ from the following data find whether there is any significant liking in habit of taking soft drinks among the categories of employees

	clerks	Teachers	officers	total
pepsi	10	25	65	100
Thumps up	15	30	65	110
fanta	50	60	30	140
Total	75	115	160	350

40.

$$E_{11} = \frac{100 \times 75}{350} = 21.42$$

$$E_{31} = \frac{140 \times 75}{350} = 30$$

$$E_{12} = \frac{100 \times 115}{350} = 32.85$$

$$E_{32} = \frac{140 \times 115}{350} = 46$$

$$E_{13} = \frac{100 \times 160}{350} = 45.71$$

$$E_{33} = \frac{140 \times 160}{350} = 64$$

$$E_{21} = \frac{110 \times 75}{350} = 23.57$$

$$E_{22} = \frac{110 \times 115}{350} = 36.14$$

$$E_{23} = \frac{110 \times 160}{350} = 50.28$$

O_{ij}	E_{ij}	$O_{ij} - E_{ij}$	$(O_{ij} - E_{ij})^2$	$\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$
10	21.42	-11.42	130.41	6.088
25	32.85	-7.85	61.62	1.875
65	45.71	20.71	428.9	9.383
15	23.57	-8.57	73.44	3.115
30	36.14	-6.14	37.69	1.042
65	50.28	15.28	233.47	0.665
50	30	20	400	13.333
60	46	-14	196	4.260
30	64	-34	1156	18.0625

② 5% Table value with 4 dof is 9.488

$$\chi^2_{Cal} > \chi^2_{table}$$

∴ we reject H_0 .

$$\chi^2_{Cal} = 60.2425$$

Correlation

- Types of correlation, coeff of correlation, properties
- Methods of finding the coeff of correlation.
- Scatter diagrams
- Direct methods - Karl Pearson's formula
Spearman's rank correlation.

Intro:

- Previously we are used to study the characteristics of only variable like marks, heights, weights, prices, ages etc..
- This type of analysis is called univariate analysis if there exists some relationship between 2 variable then, the statistical analysis of such data is called bivariate analysis.
- We may be interested to find any relation b/w 2 variables under study. For example, we can easily conclude that there is some relationship between the price of the sales, price and the production of commodity.

→ Correlation refers to the relationships b/w 2 or more variables

we know that, there exists relationship b/w the heights of father and a son, wages & price index,

The study of the relationship is called Correlation. It measures the closeness of the relationship b/w the variables.

Correlation:-

Correlation is a statistical analysis which measures and analysis the degree which 2 variables fluctuate with reference to each other.

The correlation expresses the relationship of two sets of variables upon each other. One variable may be called independent & the other is dependent.

Types of Correlation:-

It is classified into many types:-

- ① Positive & Negative
- ② Simple & Multiple
- ③ Partial & Total
- ④ Linear & Non-Linear

① Positive & Negative:-

- If 2 variables tend to move in same direction is called positive correlation.

ex:- Positive:- heights & weights

Rainfall & Yield of crops

Price & supply

If 2 variables tend to move together in opp. direction is called -ve correlation, or inverse correlation.

② Simple and Multiple correlation:-

→ When we study only 2 variables the relationship is described as simple correlation. But in a multiple correlation we may study more than 2 variables simultaneously.

③ Partial & total correlation:-

→ The study of 2 variables excluding some other variables is called partial correlation. In total correlation all the facts are taken in consideration.

④ Linear & Non-linear:-

→ If the ratio of change b/w 2 variables is uniform then there will be linear correlation between them.

→ In a non-linear correlation the amount of change of one variable does not bear a constant ratio of the amount of change in the other variables.

Methods of studying correlation:-

→ There are 2 different methods for finding out the relationship b/w 2 variables they are

① Graphical methods

② Mathematical methods.

→ The Graphical methods are scatter diagrams & simple graphs.

→ Mathematical methods are

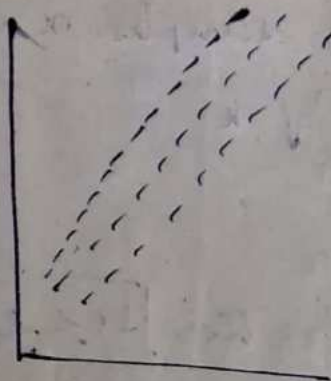
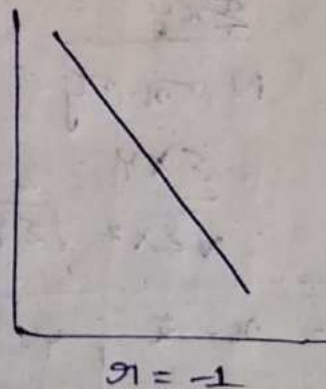
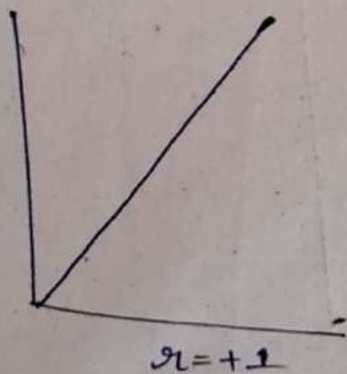
- (a) Karl's Pearson's coefficient of correlation
- (b) Spearman's rank "
- (c) Method of least squares.

Scatter diagrams:-

↳ The scatter diagram is a chart obtained by plotting 2 variables to find out whether there is any relationship b/w them.

↳ In this diagram 'X' variables are plotted on the horizontal lines & 'Y' variables are plotted on vertical lines.

↳ Various scatter diagrams are briefly shown here



② Mathematical methods:-

(a) Karl's Pearson's coefficient of correlation

↳ He is a British statistical suggested a mathematical method for measuring the magnitude of linear relationships b/w 2 variables. This is known as Pearson's coefficient of correlation and is denoted by 'r'. This method is most widely used it is also called product-movement correlation coefficient.

↳ There are several formula's to evaluate the value of 'r' they are

1) $r = \frac{\text{covariance of } xy}{\sigma_x \cdot \sigma_y}$

2) $r = \frac{\sum xy}{N \sigma_x \cdot \sigma_y}$

3) $r = \frac{\sum XY}{\sqrt{\sum X^2} \sqrt{\sum Y^2}}$

where $x = x - \bar{x}$, $y = y - \bar{y}$

where $\bar{x}$ are means of the samples x & y ,

σ_x = standard deviation of x

σ_y = S.D of y

Note:-

1) The correlation coefficient lies $-1 \leq r \leq 1$

ex: find if there is any significant correlation b/w the heights and weights given below

heights in inches ^(x)	57	59	62	63	64	65	55	58	57
weight in lbs ^(y)	113	117	126	126	130	129	111	116	112

find the correlation b/w hts & wts.

soln: The means of the given values are

$$\bar{x} = \frac{57 + 59 + 62 + 63 + 64 + 65 + 55 + 58 + 57}{9}$$

$$= 60$$

$$\bar{y} = \frac{113 + 117 + 126 + 126 + 130 + 129 + 111 + 116 + 112}{9}$$

$$= 120$$

$$\sum x^2 - \frac{(\sum x)^2}{n} = 3 + 1 + 4 + 9 + 16 + 25 + 25 + 16 + 9 = 120$$

$X = x - \bar{x}$	$Y = y - \bar{y}$	XY	X^2	Y^2
-3	-7	21	9	49
-1	-3	3	1	9
+2	6	12	4	36
3	6	18	9	36
4	10	40	16	100
5	9	45	25	81
-5	-9	45	25	81
-2	-4	8	4	16
-3	-8	24	9	64
		$\sum XY = 216$	$\sum X^2 = 102$	$\sum Y^2 = 472$

$$r = \frac{\sum XY}{\sqrt{\sum X^2} \sqrt{\sum Y^2}} = \frac{216}{\sqrt{102} \sqrt{472}} = 0.9844$$

③ Psychological test of test of Intelligence and of Engineering ability were applied to 10 students. Here is a record of ungrouped data showing Intelligence ratio and Engineering ratio. Calculate the coefficient of correlation.

I.R	105	104	102	101	100	99	98	96	93	92
ER	101	103	100	98	95	96	104	92	97	94

Soln:

$$\bar{x} = \frac{105 + 104 + 102 + 101 + \dots + 92}{10}$$

$$= 99$$

$$\bar{y} = \frac{101 + 103 + 100 + 98 + \dots + 94}{10} = \frac{980}{10} = 98$$

$X = \bar{x} - x$	$Y = \bar{y} - y$	XY	X^2	Y^2
+6	3	18	36	9
5	5	25	25	25
3	2	6	9	4
2	0	0	4	0
1	-3	-3	1	9
0	-2	0	0	4
-1	+6	-6	1	36
-3	+6	18	9	36
-6	-1	6	36	1
-7	-4	28	49	16
		92	170	140

$$r = \frac{\sum XY}{\sqrt{\sum X^2} \sqrt{\sum Y^2}} = \frac{92}{\sqrt{170} \times \sqrt{140}} = 0.59$$

6/05/25

- ③ With the following data in 6 cities calculate the coefficient of correlation by Pearson's method the density of population and the death rates

cities	Area in Sq. km	population in 1000's	no. of deaths
A	150	30	300
B	180	90	1440
C	100	40	560
D	60	42	840
E	120	72	1224
F	80	24	312

$$\bar{x} = \frac{30 + 90 + 40 + 42 + \cancel{72} + 24}{6} = \frac{298}{6}$$

→ To solve this we have to find density of population and death rates

1) The density of population = $\frac{\text{Population}}{\text{Area (sq. km)}}$

2) The death rates = $\frac{\text{No. of deaths}}{\text{population}}$

The tabulated values of density of population and the death rates are given

Density (x)	200	500	400	700	600	300
Death (rate)	10	16	14	20	17	13

$$\bar{x} = \frac{200 + 500 + 400 + \dots + 200}{6} = \frac{2700}{6} = 450$$

$$\bar{y} = \frac{10 + 16 + 14 + \dots + 13}{6} = \frac{90}{6} = 15$$

$X = x - \bar{x}$	$Y = y - \bar{y}$	XY	X^2	Y^2
-250	-5	1250	+62500	25
50	+1	50	2500	1
-50	-1	50	2500	1
250	5	1250	62500	25
150	2	300	22500	4
-150	-2	300	22500	4
		$\sum XY$ 3200	$\sum X^2$ 1,75,000	$\sum Y^2$ 60

$$r = \frac{\sum XY}{\sqrt{\sum X^2} \sqrt{\sum Y^2}}$$

$$= \frac{3200}{\sqrt{1,75,000} \sqrt{60}} = 0.987$$

When deviations are taken from assumed mean,

↳ When actual mean is not a whole number, but a fraction or when the series is too large, the calculation by direct method will involve a lot of time. To avoid such tedious calculations we can use the assumed mean method and find correlation coefficient using formula;

$$r = \frac{\sum xy - \frac{\sum x \sum y}{N}}{\sqrt{\sum x^2 - \frac{(\sum x)^2}{N}} \sqrt{\sum y^2 - \frac{(\sum y)^2}{N}}}$$

where, x = deviation of the terms of x from an assumed mean i.e. $\boxed{x = x - A}$
 $\boxed{y = y - A}$

N = total no. of items

① Calculate Karl Pearson's coefficient of correlation of the following data:

x :	28	41	40	38	35	33	40	32	36	33
y :	23	34	33	34	30	26	28	31	36	32

$$\bar{x} = 35.6 \quad \bar{y} = 31.3$$

$$A = 35, \quad A = 31$$

$X = x - A$ $= x - 35$	$Y = y - A$ $= y - 31$	XY	X^2	Y^2
-7	-8	56	49	64
6	3	18	36	9
5	2	10	25	4
3	2	6	9	9
0	-1	0	0	1
-2	-5	10	4	25
5	-8	-15	25	9
-3	0	0	9	0
1	5	5	1	25
-2	7	-14	4	49
$\Sigma X = 6$	$\Sigma Y = 3$	$\Sigma XY = 79$	$\Sigma X^2 = 162$	$\Sigma Y^2 = 195$

$$r = \frac{\Sigma XY - \frac{\Sigma X \Sigma Y}{N}}{\sqrt{\Sigma X^2 - \frac{(\Sigma X)^2}{N}} \sqrt{\Sigma Y^2 - \frac{(\Sigma Y)^2}{N}}}$$

$$r = \frac{79 - \frac{18}{10}}{\sqrt{162 - \frac{36}{10}} \sqrt{195 - \frac{9}{10}}} = \frac{77.2}{\sqrt{161.4} \sqrt{194.1}} = 0.45$$

X^2	Y^2
49	64
36	9
25	4
9	9
0	1
4	25
25	9
9	0
1	25
4	49
$\Sigma X^2 = 162$	$\Sigma Y^2 = 195$

$$\frac{(24)^2}{N}$$

$$\frac{77.2}{\sqrt{75.4} \sqrt{194.1}}$$

$$= 0.45$$

Rank correlation:-

It cannot be measured quantitatively as in the case of Pearson's coefficient correlation.

• It is based on the Ranks given to the observation.
Rank correlation is applicable only to the individual observations.

• The formula for spearman's correlation is given by

$$P = 1 - \frac{6 \Sigma D^2}{N(N^2 - 1)}$$

where P = Spearman correlation

D^2 = sum of the squares of the difference of two ranks.

N = No. of pair observation.

Note:-

- ① The value of P lies b/w -1 & 1
- ② If $P = 1$ then there is complete agreement in the order and the direction of ranks is same.
- ③ If $P = -1$ then there is complete disagreement in the order and they are in opp. direction.

- (i) Following are the ranks obtained by 10 students in two subjects statistics and mathematics to what extent the knowledge of the student of 2 subjects is reported.

statistics:	1	2	3	4	5	6	7	8	9	10
maths	2	4	1	5	3	9	7	10	6	8

Rank in statistics (x)	Rank in math (y)	D = x - y	D ²
1	2	-1	1
2	4	-2	4
3	1	2	4
4	5	-1	1
5	3	2	4
6	9	-3	9
7	7	0	0
8	10	-2	4
9	6	3	9
10	8	2	4
			40

$$P = 1 - \frac{6 \sum D^2}{N(N^2 - 1)}$$

$$= 1 - \frac{6 \times 40}{10(10^2 - 1)}$$

$$= 1 - \frac{240}{99}$$

$$= 1 - \frac{8}{33} = \frac{25}{33} = 0.7575$$

2) A random sample of 5 students is selected and their marks in maths and statistics are found to be:

	1	2	3	4	5
maths	85	60	73	40	90
statistics	93 ₁	75 ₃	65 ₄	50 ₅	80 ₂

maths	1	2	3	4	5
statistics	2	1	4	3	5

Ranks in maths (x)	Ranks in statistics (y)	D = x - y	D ²
1	2	-1	1
2	1	1	1
3	4	-1	1
4	3	1	1
5	5	0	0
			$\sum D^2 = 4$

$$P = 1 - \frac{6 \sum D^2}{N(N^2 - 1)}$$

$$= 1 - \frac{6 \times 4}{5 \times 24}$$

$$= 1 - \frac{1}{5} = \frac{4}{5} = 0.8$$

③. 10 competitors in a musical. Test were ranked by 3 judges A, B, C in the following order.

A	1	6	5	10	3	2	4	9	7	8
B	3	5	8	4	7	10	2	1	6	9
C	6	4	9	8	1	2	3	10	5	7

using rank correlation method which pair of judges the nearest approach to common likely in music.

A	B	C	D_1	D_2	D_3	D_1^2	D_2^2	D_3^2
x	y	z	$x-y$	$y-z$	$z-x$			
1	3	6	-2	-3	5	4	9	25
6	5	4	1	1	-2	1	1	4
5	8	9	-3	-1	4	9	1	16
10	4	8	6	-4	-2	36	16	4
3	7	1	-4	6	-2	16	36	4
9	10	2	-8	8	-1	64	64	1
4	2	3	2	-1	1	4	1	1
7	1	10	8	-9	-2	64	81	4
7	6	5	1	1	-2	1	1	4
8	9	7	-1	2	1	1	4	1
						<u>200</u>	<u>214</u>	<u>60</u>

$$1 - \frac{6(200)}{10(99)} = \frac{33-40}{33}$$

$$\frac{33-40}{33} = \frac{-7}{33} = -0.2121$$

$$1 - \frac{6(214)}{10(99)} = \frac{33-42}{33}$$

$$= -0.2969$$

$$1 - \frac{6(60)}{10(99)} = \frac{33-4}{33}$$

$$= 0.6363$$

AC Equal or Repeated Ranks:-

If any two or more persons bracketed equal in classification or if there is more than one then item with same values. Then the spearman formula for calculating the Rank correlation coefficient breaks down.

In this case common ranks are given to repeated items. The common rank is avg of ranks, which these items could have assumed if they are different from each others and next item we get rank next to ranks already assumed.

$$r = 1 - \frac{6 \sum D^2 + \frac{1}{12} m(m^2 - 1) + \dots}{N(N^2 - 1)}$$

m = no. of items where ranks are common.

Q From the following data calculate the Rank correlation coefficient after making adjustment for equal Ranks.

X:	48	33	40	9	16	16	25	24	16	57
Y:	13	13	24	6	15	4	20	9	6	19
Rank by X	3	5	4	10	8	8	1	6	8	2
Rank by Y	5.5	5.5	1	8.5	4	10	2	7	8.5	3
D = X - Y	-2.5	-0.5	3	1.5	4	-2	-1	-1	-0.5	-1
D ²	6.25	0.25	9	2.25	16	4	1	1	0.25	1
										$\sum D^2 = 41$

$$P = 1 - 6 \left[\frac{\sum D^2 + \frac{1}{12} m(m^2-1)}{N(N^2-1)} \right]$$

$$N = 10$$

$$P = 1 - 6 \left[\frac{41 + \frac{1}{12} \times 3(3^2-1) + \frac{1}{12} \times 2(2^2-1) + \frac{1}{12} \times 2(2^2-1)}{10(10^2-1)} \right]$$

$$= 1 - 6 \left[\frac{41 + 2 + \frac{1}{2} + \frac{1}{2}}{10 \times 99} \right]$$

$$= 1 - 6 \times \left[\frac{44}{990} \right]$$

$$P = 0.733$$

② Obtain the Rank correlation coefficient for the following data

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70
Rank by X	4	6	2.5	9	6	1	2.5	10	8	6
Rank by Y	5	7	3.5	10	1	6	3.5	9	8	2
D = x - y	-1	-1	-1	-1	5	-5	-1	1	0	4
D ²	1	1	1	1	25	25	1	1	0	16
										Σ D ² = 72

$$N = 10$$

$$\begin{aligned}
 r &= 1 - \frac{6 \left[72 + \frac{1}{12} 2(2^2-1) + \frac{1}{12} 3(3^2-1) + \frac{1}{12} 2(2^2-1) \right]}{10(10^2-1)} \\
 &= 1 - \frac{6 \left[72 + \frac{1}{2} + 2 + \frac{1}{2} \right]}{10 \times 99} \\
 &= 1 - \frac{6 \left[\frac{75}{10 \times 99} \right]}{10 \times 99} = 0.545
 \end{aligned}$$

③ A sample of 12 items gave the following data
Cal the coeff. Rank correlation

x	65	63	67	64	68	62	70	66	68	67	69	71
y	68	66	68	65	69	66	68	65	71	67	68	70
Rank by x	9	11	6.5	10	4.5	12	2	8	4.5	6.5	3	1
Rank by y	5.5	9.5	5.5	11.5	3	9.5	5.5	11.5	1	8	5.5	2
D = x - y	-3.5	-2.5	1	-1.5	1.5	2.5	-3.5	-3.5	-3.5	-1.5	-2.5	-1
D ²	12.25	6.25	1	2.25	2.25	6.25	12.25	12.25	12.25	2.25	6.25	1

$$\sum D^2 = 72.5$$

$$r = 1 - \frac{6 \left[72.5 + \frac{1}{12} \times 2(2^2-1) + \frac{1}{12} \times 2(2^2-1) + \frac{1}{12} \times 4(4^2-1) + \frac{1}{12} \times 2(2^2-1) + \frac{1}{12} \times 2(2^2-1) \right]}{12(12^2-1)}$$

$$= 1 - \frac{6 \left[72.5 + \frac{1}{2} + \frac{1}{2} + 5 + \frac{1}{2} + \frac{1}{2} \right]}{12 \times 143}$$

$$= 1 - \frac{6 \times 79.5}{12 \times 143} = 0.722$$

VI - Curve fitting and Regression analysis:



- 1) By method of least squares
- 2) fitting of a straight line, fitting of a second degree Polynomial, fitting of general curves.

Regression analysis

→ linear and multiple regression

Definitions:-

Curve fitting is the method of finding equation of a curve that approximates a given set of values (x, y) .

→ The method of least squares is the most systematic procedure to fit a unique curve to the given data values (x_i, y_i) $i = 1, 2, 3, \dots, n$.

→ Principle of least squares:

→ The sum of squares of residues should be minimum i.e. $E = d_1^2 + d_2^2 + \dots + d_n^2$ is minimum
 $d_i = \text{real value} - \text{Expected value}$.

① Fitting of a straight line:-

Let $y = a + bx$ is a straight line to be fitted to the given data (x_i, y_i) using the following steps:-

i) step-1:- Consider the set of values (x_i, y_i)

ii) find sum of x_i $\sum_{i=1}^n x_i$, $\sum_{i=1}^n y_i$, $\sum_{i=1}^n x_i^2$, $\sum_{i=1}^n x_i y_i$
 $i = 1, 2, 3, 4, \dots, n$

iii) substitute the values in the normal equations.

$$y = bx + a \quad \text{--- (1)}$$

$$\sum_{i=1}^n y_i = na + b \sum_{i=1}^n x_i \quad \text{--- (2)}$$

$$\sum_{i=1}^n x_i y_i = a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 \quad \text{--- (3)}$$

iv) Solve the above normal equations and find the values a and b

v) Subst these values of a and b in eqn (1) we get the required st. line.

3) Fit a st. line to the following data by the method of least squares

$x :$	0	5	10	15	20	25
$y :$	12	15	17	22	24	30

Soln let $y = a + bx$ --- (1)

be the st. line to be fitted to the given data

x	y	xy	x^2
0	12	0	0
5	15	75	25
10	17	170	100
15	22	330	225
20	24	480	400
25	30	750	625
$\sum x_i =$ 75	$\sum y_i =$ 120	$\sum x_i y_i =$ 1805	$\sum x_i^2 =$ 1375

The normal equations of the given st. line are

$$\begin{cases} 120 = 6a + 75b \\ 1805 = 75a + 1375b \end{cases} \text{--- (4)}$$

$$a = 11.285 \quad b = 0.697$$

Sub (a) and (b) in (1)

$$y = 11.285 + (0.697)x$$

② By the method of least squares for following data

x	0	5	10	15	20
y	7	-11	16	20	26

x	y	xy	x ²
0	7	0	0
5	-11	-55	25
10	16	160	100
15	20	300	225
20	26	520	400
$\Sigma x_i = 50$	$\Sigma y_i = 58$	$\Sigma x_i y_i = 925$	$\Sigma x_i^2 = 750$

The normal eqn of given st. line are

$$\begin{cases} 58 = 50a + 50b \\ 925 = 50a + 750b \end{cases} \text{--- (4)}$$

$$y = -2.8 + (1.38)x$$

Fitting of second degree Polynomial:-

Let $y = a + bx + cx^2$ — (1)

to be fitted into the parabola by given data

i) Consider the set of values (x_i, y_i) $i=1$ to n

(ii) Find $\sum x_i, \sum y_i, \sum x_i y_i, \sum x_i^2, \sum x_i^2 y_i, \sum x_i^3, \sum x_i^4$

subst these values in the normal equation

(iii) $\sum_{i=1}^n y_i = na + b \sum_{i=1}^n x_i + c \sum_{i=1}^n x_i^2$ — (2)

$$\sum_{i=1}^n x_i y_i = a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 + c \sum_{i=1}^n x_i^3 \quad \text{--- (3)}$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4 \quad \text{--- (4)}$$

$y = a + bx + cx^2$ — (1)

iv) solve the above normal equations to get the values a, b & c

subst these values of a, b, c in equation (1)

we get the required 2nd degree polynomial.

Ex:-

① Fit a second degree polynomial following data by the method of least squares.

x_i	1	3	5	7	9
y_i	2	7	10	11	19

x	y	xy	x^2	x^2y	x^3	x^4
1	2	2	1	2	1	1
3	7	21	9	63	27	81
5	10	50	25	250	125	625
7	11	77	49	539	343	2401
9	19	171	81	1539	729	6561
$\Sigma x_i = 25$	$\Sigma y_i = 49$	$\Sigma x_i y_i = 321$	$\Sigma x_i^2 = 165$	$\Sigma x_i^2 y_i = 2393$	$\Sigma x_i^3 = 1225$	$\Sigma x_i^4 = 9669$

$$49 = 5a + 25b + 165c \quad \text{--- (1)}$$

$$321 = 25a + 165b + 1225c$$

$$2393 = 165a + 1225b + 9669c$$

$$a = 1.5143 \quad b = 1.1857 \quad c = 0.0714$$

$$y = (1.5143) + (1.1857)x + (0.0714)x^2$$

③ Fitting of exponential curve:

Let $y = ae^{bx}$ --- (1) is an exponential curve is fitted to given data using following steps:

(i) consider, set of data values (x_i, y_i) $i = 1, \dots, n$

(ii) Apply logarithms on both sides of eqn (1)

$$\log y = \log (ae^{bx})$$

$$= \log a + \log e^{bx}$$

$$= \log a + bx \log e$$

$$\log y = \log a + bx$$

iii) Consider $y = \log y$, $A = \log a$

$$y = A + bx \quad \text{--- (2)}$$

Eqⁿ is a st. line equation.

iv) The normal eq^{ns} of st. line are

$$\sum_{i=1}^n y_i = nA + b \sum_{i=1}^n x_i \quad \text{--- (3)}$$

$$\sum_{i=1}^n x_i y_i = A \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2$$

v) find the values $\sum y_i$, $\sum x_i$, $\sum x_i y_i$, $\sum x_i^2$

vi) substitute these values in above normal eq^{ns}

solve these normal eq^{ns} we get the

unknown A & b and $a = e^A$

vii) Subst these values of a & b in eqⁿ ① we get the required exponential curve.

① Fit an exponential curve for the following data by the method of least-squares.

x	1	2	3	4	5
y	2	3	5	8	10

$y = ae^{bx}$ --- ① to be fitted to the given

data

The normal eq^{ns} of given curve are

$$\sum_{i=1}^n y_i = nA + b \sum_{i=1}^n x_i \quad \text{--- (2)} \quad \left[y = \log y \right]$$

$$\sum_{i=1}^n x_i y_i = A \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 \quad \text{--- (3)} \quad \left[\begin{array}{l} A = \log e \\ a = e^A \end{array} \right]$$

x_i	1	2	3	4	5	15
y_i	2	3	5	8	10	

$$\sum x_i = 15$$

$$Y = \log y \quad 0.693 \quad 1.098 \quad 1.669 \quad 2.079 \quad 2.302 \quad 7.7827 \quad \sum y_i$$

$$\sum x_i y_i = 0.693 \quad 2.196 \quad 4.82 \quad 8.316 \quad 11.513 \quad 27.547 = \sum x_i y_i$$

$$\sum x_i^2 = 1 \quad 4 \quad 9 \quad 16 \quad 25 \quad 55 = \sum x_i^2$$

$$7.7827 = 5A + 15B$$

$$27.547 = 15A + 55B$$

$$A = 0.29687$$

$$B = 0.41989$$

$$a = 1.3456$$

$$y = 1.3456 e^{0.4199x}$$

* Fitting of Power curves:-

let $y = ab^x$ — (1) is the power curve to be fitted to given data

apply log on B.S.

$$\log y = \log (ab^x)$$

$$= \log a + \log b^x$$

$$\log y = \log a + x \log b$$

$$Y = \log y, \quad A = \log a, \quad B = \log b$$

$$Y = A + Bx \quad \text{--- (2)}$$

eqn (2) is a line

The Normal eqns of st-line are

$$\sum y_i = nA + B \sum x_i \quad \text{--- (3)}$$

$$\sum x_i y_i = A \sum x_i + B \sum x_i^2 \quad \text{--- (4)}$$

find $\sum y_i$, $\sum x_i$, $\sum x_i y_i$, $\sum x_i^2$

using the given data

subl these values in eqn (3) & (4) we get

The unknown A & B but $\boxed{a=e^A, b=e^B}$

subl a & b in eqn (1) we get power curve.

* The power eqn is $y = ax^b$ --- (1)

$$\log y = \log(ax^b)$$

$$= \log a + b \log x$$

$$Y = \log y, X = \log x, A = \log a$$

$$Y = A + BX \quad \text{--- (2)}$$

eqn (2) in st-line

Normal eqn of st-line are

$$\sum y_i = nA + b \sum x_i \quad \text{--- (3)}$$

$$\sum x_i y_i = A \sum x_i + b \sum x_i^2 \quad \text{--- (4)}$$

find $\sum x_i$, $\sum y_i$, $\sum x_i y_i$, $\sum x_i^2$

$$\boxed{a=e^A}$$

① fit the power curves $y = ab^x$, $y = ax^b$ for the following:

x	1	2	3	4	5
y	2	3	5	8	10

Soln

$$y = ab^x \text{ --- ①}$$

x	0	0.693	1.098	1.3862	1.6094	1.7866
X	1	2	3	4	5	15
Y	2	3	5	8	10	-

$$\log y \quad 0.693 \quad 1.098 \quad 1.604 \quad 2.079 \quad 2.302 \quad 7.7829$$

$$x^2 \quad 1 \quad 4 \quad 9 \quad 16 \quad 25 \quad 55$$

$$x_1 y_1 \quad 0.693 \quad 2.196 \quad 4.8282 \quad 8.316 \quad 11.5113 \quad 27.547$$

$$x_1^2 \quad 0 \quad 0.48 \quad 1.188 \quad 1.904 \quad 2.590 \quad 6.162$$

$$7.7829 = 5A + B15$$

$$27.547 = A15 + B55$$

$$A = 0.29687$$

$$B = 0.41987$$

$$a = 1.3456$$

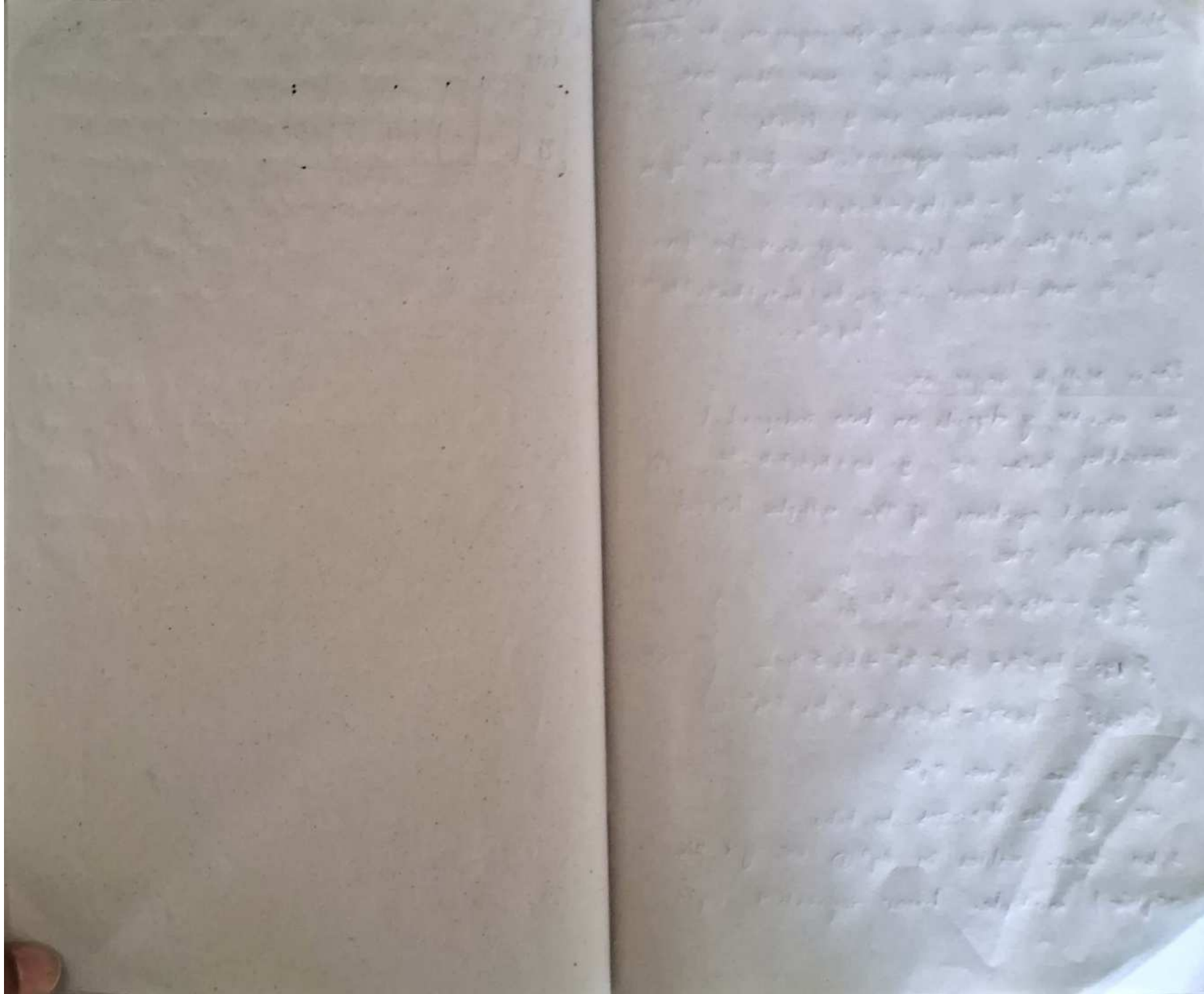
$$b = 1.5217$$

$$y = (1.345) (1.5217)^x$$

Q. Fit a power curves $y = ax^b$ for the following

data.

x:	1	2	4	5	6
y:	1	5	10	22	38



17/06/2024
Multiple regression: In multiple regression the dependent variable y is a funⁿ of more than one independent variable i.e $y = f(x_1, x_2, \dots)$

- In multiple linear regression, the function ' f ' is linear i.e $y = b_0 + b_1x_1 + b_2x_2 + \dots$

→ In multiple non-linear regression, the funⁿ ' f ' is non-linear i.e $y = b_0 + b_1x_1 + b_2x_2 + b_3x_1x_2 + b_4x_1^2 + \dots$

Linear Multiple regression:

the variable y depends on two independent variables x_1, x_2 i.e $y = b_0 + b_1x_1 + b_2x_2$ — (1)

The normal equations of the multiple linear regression are

$$\sum_{i=1}^n y_i = nb_0 + b_1 \sum_{i=1}^n x_{1i} + b_2 \sum_{i=1}^n x_{2i}$$

$$\sum_{i=1}^n x_{1i} y_i = b_0 \sum_{i=1}^n x_{1i} + b_1 \sum_{i=1}^n x_{1i}^2 + b_2 \sum_{i=1}^n x_{1i} x_{2i}$$

$$\sum_{i=1}^n x_{2i} y_i = b_0 \sum_{i=1}^n x_{2i} + b_1 \sum_{i=1}^n x_{1i} x_{2i} + b_2 \sum_{i=1}^n x_{2i}^2$$

Solving the above eq^s

we get the ⁿknowns b_0, b_1, b_2

subst these values in eqⁿ (1) we get the required multiple linear regression eqⁿ.

① find the multiple linear regression equation of x_1 on x_2 and x_3 from the given data

x_1	2	4	6	8
x_2	3	5	7	9
x_3	4	6	8	10

a. The Linear multiple regression equation is

$$x_1 = b_0 + b_1 x_2 + b_2 x_3$$

①

$$x_1 = 2x_2 - x_3$$

The normal equations are

$$\sum x_1 = nb_0 + b_1 \sum x_2 + b_2 \sum x_3$$

$$\sum x_1 x_2 = b_0 \sum x_2 + b_1 \sum x_2^2 + b_2 \sum x_2 x_3$$

$$\sum x_1 x_3 = b_0 \sum x_3 + b_1 \sum x_2 x_3 + b_2 \sum x_3^2$$

②

x_1	x_2	x_3	$x_1 x_2$	$x_1 x_3$	$x_2 x_3$	x_2^2	x_3^2
2	3	4	6	8	12	9	16
4	5	6	20	24	30	25	36
6	7	8	42	48	56	49	64
8	9	10	72	80	90	81	100
$\sum x_1 = 20$	$\sum x_2 = 24$	$\sum x_3 = 28$	$\sum x_1 x_2 = 140$	160	188	164	216

$$20 = 14b_0 + 24b_1 + 28b_2$$

$$140 = 24b_0 + 164b_1 + 188b_2$$

$$160 = 28b_0 + 188b_1 + 216b_2$$

$$b_0 = 0, b_1 = 2, b_2 = -1$$